MILITARY SPECIFICATION

GENERATOR PULSE ASSEMBLY: 10556650

1. SCOPE

1.1 This specification covers a generator pulse assembly which is a part of the Test Set, Radar AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18 Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWING

U.S. Army, Frankford Arsenal

F10556650	Pulse Generator Assembly

PACKAGING DATA SHEET

10556650	Packaging of Pulse Generator Assembly

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(Copies of specifications, standards and drawings required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The pulse generator assembly shall be fabricated in accordance with Drawing F10556650 and drawings pertaining thereto:

3.2 General specifications.- Where applicable the pulse generator shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The pulse generator assembly shall be mounted on the vibration table in normal service position and shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

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3.3.2 Shock.- The pulse generator assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating.- The pulse generator assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating.- The pulse generator assembly, shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating.- The pulse generator assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +131°F.

3.3.4.2 Nonoperating (storage).- The pulse generator assembly shall not suffer damage when subjected to the temperature extremes of -80°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating.- The pulse generator assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.- Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance.

3.5.1 Power requirements.

(a) A potential of + 15 Vdc ± 5% is applied to E9
(b) E11 and E3 are grounded
(c) There is a resistance variable between 0 ohms & 50 k connected between E1 and E2.
(d) A resistance of 100 ohms is connected between E5 and E6
(e) E10 is connected to a switch which provides ground potential in the NC position and + 15 ± 5% Vdc in the NO position

3.5.2 E5 signal output.- With conditions as stated in 3.5.1 and a 100 ohm load between E5 and ground there shall be a pulse signal of 5 ± 1V at E5 variable in PRF with adjustment of resistance between E1 and E2.

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3.5.3 E8 driver output.

3.5.3.1 With conditions as stated in 3.5.1 a 100 ohm load between E8 and ground, and a ground potential applied to E10, there shall be a pulse signal of 5 ± 1V at E8 variable in PRF with adjustment of resistance between E1 and E2.

3.5.3.2 With conditions as stated in 3.5.1 and 15 Vdc ± 5% applied to E10 there shall be no pulse signal at E8.

3.5.4 PRF adjust resistance.

3.5.4.1 With the resistance between E1 and E2 set at 50k and conditions as stated in 3.5.2, the PRF of the pulse at E5 shall not be greater than 4KHz.

3.5.4.2 With conditions as stated in 3.5.2 and the resistance between E1 and E2 set at 0 ohms, the PRF of the pulse at E5 shall be not less than 26 KHz.

3.5.5 Width adjust R8.- R8 shall be adjusted so that the pulse width of the pulses at E8 and E5 is .3 ± .1 microsecond.

3.5.6 With conditions as stated in 3.5.1, E2 connected to E3 and a pulse of between 5 V and 20 V with a PRF of between 10 Hz and 90KHz and a pulse width of .4 ± .1 microsecond applied to E4 there shall be a 5 ± 1 V pulse at E5 with pulse width of .3 ± .1 microsecond and a PRF equal to that of the input pulse.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one unit manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: None		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Power requirements	3.5.1	4.6.3.1
103. E5 - Signal Output	3.5.2	4.6.3.2
104. E8 - Driver Output	3.5.3	4.6.3.3
105. PRF - Adjust Resistance	3.5.4	4.6.3.4
106. Width Adjust - R8	3.5.5	4.6.3.5
107. External Sync E4	3.5.6	4.6.3.6
MINOR: None		

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One pulse generator shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table I prior to the examination of Table II.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawing - visual
302. General specification	3.3	MIL-F-13926 - visual
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude (operating)	3.3.3.1	4.6.1.3
306. Altitude (nonoperating)	3.3.3.2	4.6.1.3
307. Temperature (operating)	3.3.4.1	4.6.1.4
308. Temperature (nonoperating)	3.3.4.2	4.6.1.5
309. Humidity (nonoperating)	3.3.5	4.6.1.6

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

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4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment.- In conjunction with 4.5.2, the test equipment listed below may be utilized in performing the applicable tests specified in 4.6. (The test arrangement of Figure 1 is suggested.)

Test Equipment

No.	ITEM	DESCRIPTION
1.	Power Supply	15 Vdc ± 5% 1 amp
2.	Switch	Single pole, 2 position
3.	Switch	Single pole 2 position
4.	Switch	Double pole, 2 position
5.	Potentiometer	50 K ± 5% ten turn
6.	Pulse Generator	PRF 10 Hz - 90 KHz, pulse width .4 ± .2 u sec, amplitude 4V to 20V
7.	Coaxial Cable	RG 180 as required
8.	Termination	100 ohm ± 5%
9.	Oscilloscope	Tektronix 454 or equivalent
10.	Multimeter	0-50 V ± 20%, full scale

4.6 Test methods and procedures.

4.6.1 Environmental.- The following environmental tests shall be performed utilizing the test conditions and facilities specified in MIL-STD-810 and MIL-F-13926 as applicable.

4.6.1.1 Vibration.- This test shall be performed as specified in MIL-STD-810, Method 514, Procedure I, with the following exceptions: the amplitude, frequency, acceleration and duration shall be as specified in 3.3.1. At the conclusion of this test, the pulse generator shall be inspected by visual and tactile means for damage, then successfully tested for compliance with the performance requirements specified.

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4.6.1.2 Shock.- This test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV, with acceleration, duration, number of shocks and direction as specified in 3.3.2. At the conclusion of this test, the pulse generator shall be inspected by visual and tactile means for evidence of damage, then successfully tested for compliance with the performance requirements specified.

4.6.1.3 Altitude - non-operating and operating.- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Operating temperatures.- This test shall be performed as specified in MIL-F-13926, Procedure II, at the temperatures specified in 3.3.4.1. While thermally stabilized at the low and high temperatures, the pulse generator shall be successfully tested for compliance with performance requirements specified.

4.6.1.5 Storage temperatures.- This test shall be performed as specified in MIL-F-13926, Procedure I, at the extreme temperatures specified in 3.3.4.2. Upon completion of the above exposures, the pulse generator shall be thermally stabilized at standard ambient temperature, then inspected by visual and tactile means for any damage or deterioration. After the above tests have been completed, the pulse generator shall be successfully tested for compliance with the performance requirements specified.

4.6.1.6 Humidity - non-operating.- This test shall be performed as specified in MIL-F-13926, Procedure I, at the relative humidity and duration specified in 3.3.5. At the completion of the above, the pulse generator shall be successfully tested for the performance requirements specified.

4.6.2 Electrical.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance.

4.6.3.1 Power requirements.- Connect the pulse generator as specified in 3.5.1 and Figure 1. With the multimeter, Item 11 measure the following voltages:

From (+)	To (-)	VDC
P1-6	P1-9	15 ± 5%
P1-10	P1-9	12 ± 5%
P1-5	P1-9	15 ± 5%

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(a) Check that the test points P1-4, P1-8, P1-3 and P1-9 are at ground potential.

(b) Unless otherwise specified, the test equipment shall be connected to the pulse generator as specified herein or as suggested in Figure 1.

4.6.3.2 E5 signal output.- Connect the potentiometer, Item 5, between P1-1 and P1-2. Connect the 100 ohm load, Item 8, between P1-A2 and P1-4. Connect the oscilloscope across the 100 ohm load. Apply the 15 volt power supply to P1-6. Vary the potentiometer adjustment and observe the pulse on the screen. The pulse amplitude shall be 5 ± 1 volts and the pulse rate frequency (PRF) shall vary between 4 KHz and 26 KHz. Disconnect the power supply.

4.6.3.3 E8 driver output.- The potentiometer shall remain connected as specified above. The 100 ohm load shall be connected between P1-A3 and ground. The oscilloscope shall be connected across the load. Re-apply the 15 volts to P1-6. Vary the potentiometer and observe the pulse. The pulse amplitude shall be 5 ± 1 volts and the PRF shall vary between 4 KHz and 26 KHz.

(a) Apply the 15 volts to P1-7 and ground and observe the oscilloscope screen. There shall be no pulse signal visible. Disconnect the power supply.

4.6.3.4 PRF adjust resistance.- With the pulse generator connected as specified in 4.6.3.2, adjust the potentiometer at P1-1 and P1-2 to zero ohms. Re-apply the 15 volts to P1-6. Observe the oscilloscope screen. The PRF shown on the screen shall not be less than 26 KHz.

(a) Adjust the potentiometer to 50 K ohms. The PRF pulse shall not be greater than 4 KHz. Disconnect the power supply.

4.6.3.5 Width adjust - R8.- Connect the oscilloscope across the 100 ohm load at P1-A3. Energize the power supply. Observe the pulse width on the screen. The pulse width shall be 0.3 ± 0.1 microsend. Adjust Resistor R8 on circuit board assembly, if necessary, to produce the above pulse width. Disconnect the power supply.

4.6.3.6 External Sync E4.- Connect the potentiometer connection P1-2 to P1-3. Connect the oscilloscope across the 100 ohm load to P1-A2. Connect the power supply to P1-6. Connect the pulse generator, Item 6 to P1-A1. Activate the power supply and pulse generator. The input pulse signal shall be between 10 Hz and 90 KHz with a pulse width 0.4 ± 0.1 usec and an amplitude between 5 volts and 20 volts. The output signals measured on the oscilloscope across the 100 load at P1-A1 and P1-A3 shall be equal to the input signals.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10556650. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The pulse generator assembly is part of Radar Test Set, AN/TPM-22.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable packaging data sheet number (see 5.1).
c. Selection of applicable level of preservation, packaging and packing.
d. Level of protection required.
e. Applicable stock number.
f. Requirement for first article submission.

Custodian:

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Preparing activity:

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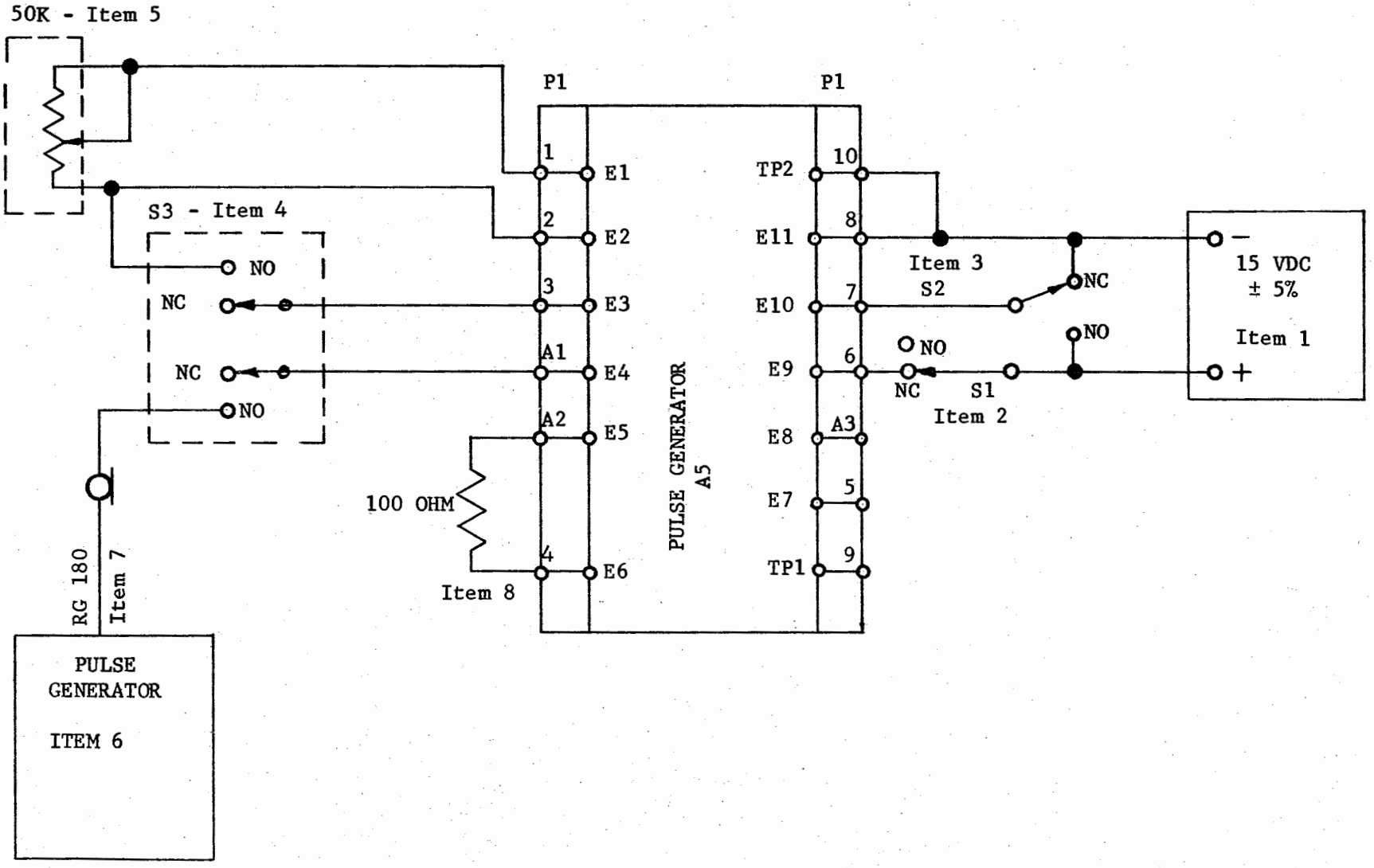


FIGURE 1

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